

Data Collection and Preprocessing Phase

Date	27 June 2025
Student ID	SWUID20250181505
Project Title	online payments fraud detection using machine learning
Maximum Marks	2 Marks

Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

Data Source	Data Quality Issue	Severity	Resolution Plan
Kaggle Dataset - PS_20174392719_1491204439457_log.csv	High Class Imbalance: The dataset is highly imbalanced, with 1,047,433 non-fraudulent transactions (99.89%) and only 1,142 fraudulent transactions (0.11%), which can bias the model towards the majority class.	High	Downsample the non-fraudulent class to 1,288 records to create a balanced dataset of 2,430 records (1,142 fraudulent, 1,288 non-fraudulent). Use random sampling with a fixed seed (<code>random_state=42</code>) to ensure reproducibility. Alternatively, explore oversampling techniques like SMOTE in future iterations.

	Irrelevant Features: The nameOrig and nameDest columns are identifiers (e.g., customer IDs) with no predictive value for fraud detection, increasing computational overhead.	Moderate	Drop the nameOrig and nameDest columns using <code>df.drop(['nameOrig', 'nameDest'], axis=1)</code> to reduce dataset dimensionality and focus on relevant features like step, type, amount, and balance-related columns.
	Categorical Data: The type column (e.g., CASH_IN, CASH_OUT, TRANSFER) is categorical, requiring transformation for machine learning models.	Moderate	Encode the type column using LabelEncoder from scikit-learn: <code>le = LabelEncoder(); df['type'] = le.fit_transform(df['type'])</code>. This converts categorical values to numerical indices (e.g., CASH_OUT=1, TRANSFER=4) for model compatibility.
	No Missing Values: The dataset has no missing values across all	Low	No action required, as the dataset is complete. Continue monitoring

	columns, as confirmed by <code>df.isnull().sum()</code> .		for missing values in future data updates to ensure consistency.
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